Dr. Eng. Ramiro M. Irastorza

Independent Researcher – CONICET
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Education

- **Ph.D. in Electronic Engineering**, National University of La Plata (UNLP), Argentina, 2010.
  Dissertation: *System Identification: Application to the In Vitro Evaluation of Bone Quality in Human Trabecular Tissue.*
- **Electronic Engineer**, National University of La Plata (UNLP), Argentina, 2005.

Current Positions

- **Independent Researcher**, CONICET (2026–Present).
- **Full Professor**, Automation Laboratory, Mechanical Engineering Department, UTN-FRLP (2025–Present).
- Vice Director, Center for Experimental and Computational Mechanics (CIMEC), UTN-FRLP.

Research Interests

- Computational bioengineering
- Finite element modeling
- Microwave tomography and inverse problems
- Radiofrequency and microwave medical therapies
- Biomedical image reconstruction
- Thermal and dielectric properties of biological tissues
- Scientific computing and machine learning

Research Funding and Leadership

- Principal Investigator of the UTN project: *Biomedical and Biomimetic Problems Solved Using Finite Elements* (2023–2026).

- Principal Investigator of the national research project: *Microwave Tomography: Reconstruction Algorithms, Experimental Validation and Applications* funded by the National Agency for Scientific and Technological Promotion (2022–2024).
- Co-Principal Investigator of the CONICET project: *Inverse Problems in Biomedical Engineering Using Sparse Representations and Artificial Intelligence* (2023–2024).

Selected International Research Project Participation

- **Improvement of Advanced Ablative Therapies through the Control of Tissue and Cellular Behavior Using Electromagnetic Fields**. Participants: Polytechnic University of Valencia, Neuroscience and Complex Systems Studies Center (CONICET), Institute of Electronics, Signal Processing and Control Research (LEICI-CONICET), and IFLySiB-CONICET. Funded by the Ministry of Economy and Competitiveness, Spain (2023–2025).
- **Radiofrequency- and Microwave-Based Technologies for Minimally Invasive Surgery**. Participants: Polytechnic University of Valencia and IFLySiB-CONICET. Funded by the Ministry of Economy and Competitiveness, Spain (2015–2017).
- **Mise en place d'un laboratoire international pour la recherche sur les matériaux structures recouvrements pour applications médicales**. Participants: Université Laval, INTI, and the Biomechanics Laboratory, School of Engineering, University of Buenos Aires. Funded by the Agence Universitaire de la Francophonie (AUF) (2010–2012).

Graduate Teaching

- **Finite Element Method with Open-Source Software** (UTN-FRLP)
- **Computational Tools for Scientists** (UNLP / UTN)
- **Statistical Methods** (UNAJ)
- **Applied Mathematics Complements** (UNAJ)
- **Radiofrequency and Biological Tissues** (Polytechnic University of Valencia, Spain)

Human Resources Training

- Supervision and co-supervision of Ph.D. students in biomedical engineering, computational mechanics, and microwave imaging.
- Advisor of undergraduate and graduate theses at UTN, UNAJ, and UNLP.
- Supervisor of CONICET and national research fellowship recipients.

Selected Publications

1. Berjano E., **Irastorza R.M.**, *Positioning of the dispersive electrode and its effect on the safety and efficacy of radiofrequency ablation*, EP Europace, 2025.

2. Collavini S., Pérez J.J., Berjano E., Fernández-Corazza M., Oddo S., **Irastorza R.M.**, *Impact of surrounding tissue-type and peri-electrode gap in stereoelectroencephalography guided (SEEG) radiofrequency thermocoagulation (RF-TC): a computational study*, International Journal of Hyperthermia, 2024.

3. Cervantes M.J., Basiuk L.O., González-Suárez A., Carlevaro C.M., **Irastorza R.M.**, *Low-Frequency Electrical Conductivity of Trabecular Bone: Insights from In Silico Modeling*, Mathematics, 2023.

4. **Irastorza R.M.**, Maher T., Barkagan M., Liubasuskas R., Berjano E., d'Avila A., *Anterior vs. posterior position of dispersive patch during radiofrequency catheter ablation: insights from in silico modelling*, EP Europace, 2023.

5. González-Suárez A., Pérez J.J., **Irastorza R.M.**, D'Avila A., Berjano E., *Computer modeling of radiofrequency cardiac ablation: 30 years of bioengineering research*, Computer Methods and Programs in Biomedicine, 2022.

6. **Irastorza R.M.**, Bovaira M., García-Vitoria C., Muñoz V., Berjano E., *Effect of the relative position of electrode and stellate ganglion during thermal radiofrequency ablation*, International Journal of Hyperthermia, 2021.

7. Fajardo J.E., Lotto F.P., Vericat F., Carlevaro C.M., **Irastorza R.M.**, *Microwave Tomography with Phaseless Data on the Calcaneus by Means of Artificial Neural Networks*, Medical & Biological Engineering & Computing, 2019.

8. **Irastorza R.M.**, d'Avila A., Berjano E., *Thermal latency adds to lesion depth after application of high-power short-duration radiofrequency energy*, Journal of Cardiovascular Electrophysiology, 2018.

9. **Irastorza R.M.**, Trujillo M., Berjano E., *How coagulation zone size is underestimated in computer modeling of RF ablation by ignoring the cooling phase*, International Journal for Numerical Methods in Biomedical Engineering, 2017.

10. **Irastorza R.M.**, Blangino E., Carlevaro C.M., Vericat F., *Modeling of the dielectric properties of trabecular bone samples at microwave frequency*, Medical & Biological Engineering & Computing, 2014.

Professional Activities

- Reviewer for international journals including:
  - International Journal of Hyperthermia
  - Medical Engineering & Physics
  - Progress In Electromagnetics Research
  - International Journal for Numerical Methods in Biomedical Engineering
- Guest Editor: *Biophysical- and Engineering-Oriented Approaches to Energy-Based Cardiac Ablation Techniques*, Journal of Cardiovascular Development and Disease.
- Co-Coordinator of the Information Technology, Communications and Electronics Area, National Agency for Scientific and Technological Promotion FONCyT (PICT calls 2022–2025).
- Organizer of PyDay La Plata (2022–2025).